A HELICAL BRANCH MODEL FOR PREDICTING ORDINATES OF ZETA ZEROS AT ALPHA 3.43397596

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Remark 0.1. This article is a research note describing a numerical and heuristic procedure. Standard facts about the Riemann zeta function are cited from the literature. The helical model, the modular branch reconstruction, and the reported numerical fits are original computational observations in the present experiment. They are not presented as theorems about the zeta function.

1. Introduction

Let $\rho_k = \beta_k + i\gamma_k$ denote the nontrivial zeros of the Riemann zeta function ordered by increasing positive ordinate. The analytic theory of these zeros is standard; see Riemann's memoir [1], the monographs of Titchmarsh and Edwards [2, 3], and large-scale numerical work such as Odlyzko's computations [5]. The present note does not attempt to prove statements about the distribution of the γ_k. Instead, it records a numerical model for reconstructing γ_k from a helical fit to a complex walk.

The starting point was the family of exponential sums

$$S_k(\alpha) = \sum_{n=1}^{k} \exp(i\alpha\gamma_n). \tag{1.1}$$

The classical theorem of Landau, in one of its standard forms, describes the large-T behaviour of sums over zeros weighted by x^ρ; in particular, when x is a prime power a main term involving the von Mangoldt function appears, while away from such arithmetic resonances the corresponding main term is absent. We use this only as standard motivation; see [2, 3]. Since

$$x^{i\gamma_n} = \exp(i(\log x)\gamma_n), \tag{1.2}$$

the frequencies $\alpha = \log x$ are natural reference values for (1.1).

The value

$$\log 31 = 3.433987204485146\ldots \tag{1.3}$$

is extremely close to the frequency used in this article,

$$3.43397596 - \log 31 = -1.1244485146 \cdot 10^{-5}. \tag{1.4}$$

Empirically, the exponential sum at the prime frequency $\log 31$ displays a degenerate arithmetic drift, while at the nearby but distinct value $\alpha = 3.43397596$ an unexpectedly coherent helical trajectory appears. This near-resonant behaviour is not, to the author's knowledge, described in the standard statements of Landau's theorem: the theorem identifies the arithmetic resonant main term, but it does not characterize the geometry of nearby off-resonant exponential walks.

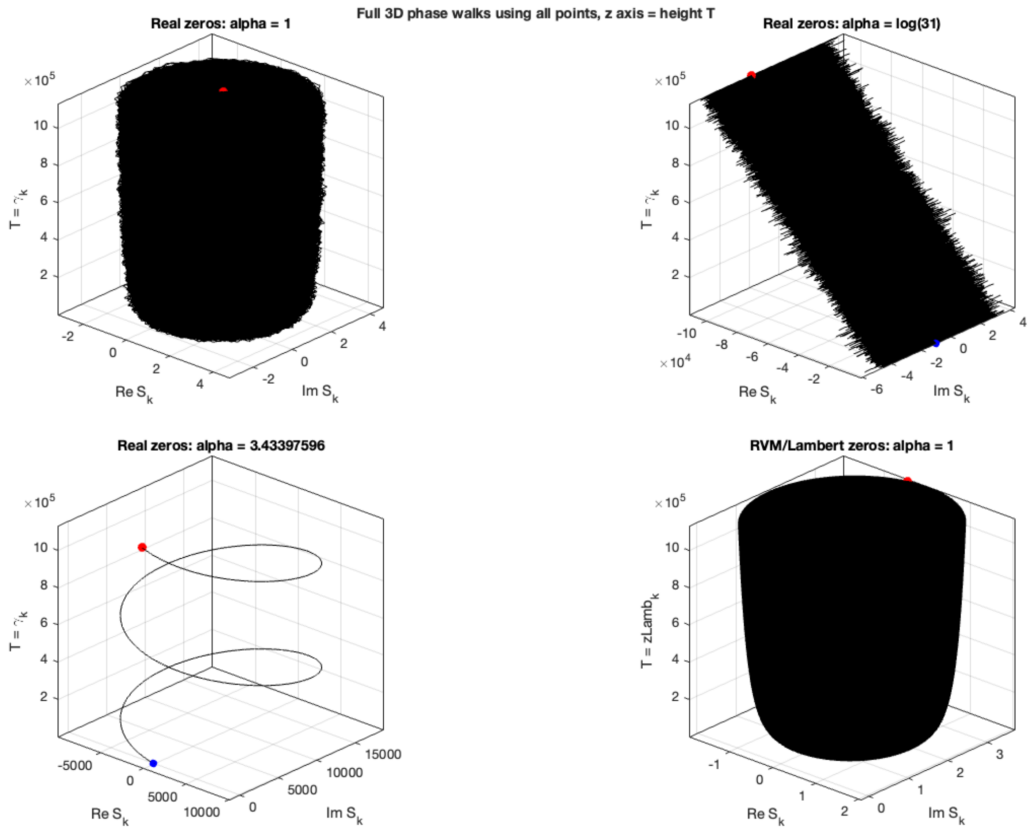


FIGURE 1. Full three-dimensional phase walks using all available points, with height on the vertical axis. The first three panels use the actual zero ordinates from `zeross.bin` at $\alpha = 1$, $\alpha = \log 31$, and $\alpha = 3.43397596$. The fourth panel replaces the zero ordinates by the smooth Riemann–von Mangoldt/Lambert heights at $\alpha = 1$.

After this helical trajectory was observed, the numerical search was organized around minimizing the apparent wall thickness of the helical tube in the near-resonant region. The value (1.5) is the result of that wall-thickness search. The purpose of the present note is to characterize the resulting helix, to make the necessary branch structure explicit, and to measure the prediction error for individual zero ordinates.

An earlier exploratory fit used only the first 100,000 zeros. That fit could be extrapolated to the full two-million-zero range, but it was outside the range used in the fit. Repeating the fit with

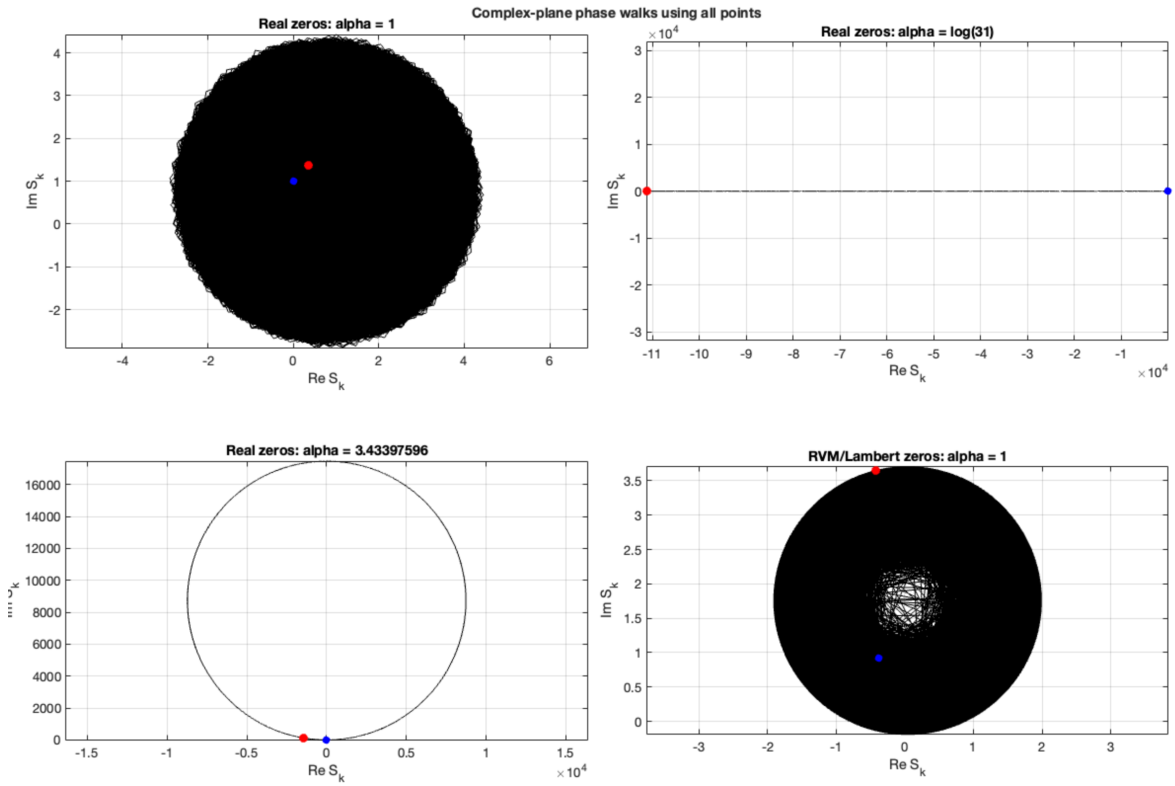


FIGURE 2. Complex-plane projections of the same four walks. The panels show the disk-like walk at $\alpha = 1$, the nearly one-dimensional arithmetic degeneration at $\log 31$, the circular projection of the optimized helical walk, and the corresponding Riemann–von Mangoldt/Lambert comparison at $\alpha = 1$.

all 2,001,052 zeros changed the helical parameters substantially, but did not materially improve the height-equation error statistics in this interval. For example, with diagnostic real branches in the height equation, the 100,000-zero fit gave a last-zero error of approximately -0.8055788970, whereas the full-set fit gave 0.4857182777; the corresponding mean absolute errors were both about 0.457. This suggests that, at least over the tested interval, the helical structure is stable enough that the main limitation is not simply the number of zeros used in the fit.

The experiment fixes

$$\alpha = 3.43397596 \tag{1.5}$$

and considers the complex walk

$$S_k = \sum_{n=1}^{k} \exp(i\alpha\gamma_n). \tag{1.6}$$

The dataset used in the computation contains

$$N = 2{,}001{,}052 \tag{1.7}$$

positive ordinates, read from the binary file `zeross.bin`.

Remark 1.1. The construction below is empirical. The fitted helix and branch counters are reported as a compact description of the observed data. The procedure is not a replacement for standard asymptotic formulae for zero counting, such as those discussed in classical references [2, 3].

2. The Helical Fit

The walk (1.6) was fitted by a six-parameter helical curve in the complex plane:

$$H(k) = C + Ak^B \exp\big(i(\omega k + \theta_0)\big). \tag{2.1}$$

The fitted parameters were

$$C = 184.581604754753 + 8696.109422755926\, i, \tag{2.2}$$

$$A = 9709.979940513593, \tag{2.3}$$

$$B = -0.007861904817, \tag{2.4}$$

$$\omega = -9.549822747506 \cdot 10^{-6}, \tag{2.5}$$

$$\theta_0 = -1.975733093439. \tag{2.6}$$

Thus the fitted helix is

$$\begin{aligned} H(k) &= (184.581604754753 + 8696.109422755926\, i) \\ &\quad + 9709.979940513593\, k^{-0.007861904817} \exp\big(i(-9.549822747506 \cdot 10^{-6} k - 1.975733093439)\big). \end{aligned} \tag{2.7}$$

Experiment 2.1. The parameters in (2.2)–(2.6) were obtained by centering the walk (1.6), unwrapping the observed argument, and fitting a linear angular speed together with a power law for the radius. This is a numerical fitting procedure, not a theorem.

3. Incremental Reconstruction of Ordinates

The first reconstruction is incremental. It writes the predicted ordinates as

$$\gamma_{\mathrm{pred}}(1) = \gamma_1, \tag{3.1}$$

and, for $k \geq 2$,

$$\gamma_{\mathrm{pred}}(k) = \gamma_1 + \sum_{n=1}^{k-1} \left(\Delta_h(n) + \frac{2\pi}{\alpha} m(n) \right). \tag{3.2}$$

Here $m(n)$ is the integer branch increment between γ_n and γ_{n+1}. The numerical value of the branch period is

$$\frac{2\pi}{\alpha} = 1.8297115007117248\ldots. \tag{3.3}$$

The smooth helical increment must be written with a lifted complex argument:

$$\Delta_h(n) = \frac{1}{\alpha} \left(\omega + \mathrm{Arg}_{\mathrm{lift}} \left(\omega + i \frac{B}{n} \right) \right). \tag{3.4}$$

This replaces the monovalued expression

$$\frac{1}{\alpha} \left(\omega + \arctan\left(\frac{B}{\omega n} \right) \right), \tag{3.5}$$

which does not determine the quadrant.

Warning 3.1. In the present fit $B < 0$ and $\omega < 0$. Therefore the complex number $\omega + iB/n$ lies in the third quadrant. A scalar arctangent cannot determine this quadrant. In computation the principal quadrant-corrected argument is computed by

$$\phi(n) = \mathrm{atan2}(B/n, \omega). \tag{3.6}$$

The lifted branch must then be chosen coherently.

4. Natural and Provoked Branch Increments

The branch increment is decomposed as

$$m(n) = m_{\mathrm{nat}}(n) + m_{\mathrm{prov}}(n), \tag{4.1}$$

where

$$m_{\mathrm{nat}}(n) \in \{0, 1\} \tag{4.2}$$

is the natural continuation of the helical branch. The cases $m_{\mathrm{nat}}(n) = 1$ are ordinary one-turn branch continuations and are not treated as failures of the helical model.

The provoked jumps are the excess turns:

$$m_{\mathrm{prov}}(n) = \sum_{r=1}^{17} \eta(r) \mathbf{1}_{n=j(r)}. \tag{4.3}$$

Here $j(r)$ is the gap index at which a modular event occurs and $\eta(r)$ is the excess over the natural one-turn continuation.

Definition 4.1. A gap index n is called provoked in this experiment if $m(n) > 1$. Its excess is

$$\eta(n) = m(n) - 1. \tag{4.4}$$

5. The Modular Model for the Branch Events

The branch-event table is represented by a Newton-binomial polynomial over the prime field with

$$p = 1{,}000{,}003. \tag{5.1}$$

For $r = 1, \ldots, 17$, define

$$j(r) \equiv \sum_{q=0}^{16} A_q \binom{r-1}{q} \pmod{p}, \tag{5.2}$$

$$e(r) \equiv \sum_{q=0}^{16} B_q \binom{r-1}{q} \pmod{p}. \tag{5.3}$$

The residues lie in the relevant integer range, so the modular residues are read as exact integer values.

The exact verification of the modular reconstruction was

$$\max_r |j_{\mathrm{model}}(r) - j_{\mathrm{data}}(r)| = 0, \tag{5.4}$$

$$\max_r |e_{\mathrm{model}}(r) - e_{\mathrm{data}}(r)| = 0. \tag{5.5}$$

The first and last events were

$$(r, j(r), e(r)) = (1, 1, 3), \tag{5.6}$$

$$(r, j(r), e(r)) = (17, 126, 1). \tag{5.7}$$

Remark 5.1. Equations (5.2) and (5.3) are an exact finite encoding of the observed branch-event table. They should not be interpreted as a proven formula for zeta zeros beyond the tested range.

6. Useful Provoked Jumps

For $\alpha = 3.43397596$, the modular table was rebuilt directly from the useful provoked events. Thus every event in the table satisfies $m(n) > 1$. The experiment found

$$\#\{n : m(n) > 1\} = 17. \tag{6.1}$$

All of these 17 provoked jumps are represented by the modular event model:

$$\{n : m(n) > 1\} = \{j(r) : 1 \le r \le 17\}. \tag{6.2}$$

The useful provoked jumps are listed in Table 1.

Table 1. Provoked jumps. The column $m(n)$ is the total branch multiplier, and $m(n) - 1$ is the excess over the natural continuation.

n	$m(n)$	$m(n) - 1$
1	4	3
2	2	1
3	3	2
5	2	1
6	2	1
8	2	1
11	2	1
14	2	1
18	2	1
21	2	1
25	2	1
33	2	1
41	2	1
46	2	1
64	2	1
79	2	1
126	2	1

The last provoked jump is

$$n = 126. \tag{6.3}$$

Consequently, for $n > 126$,

$$m_{\mathrm{prov}}(n) = 0. \tag{6.4}$$

After this point the model continues with the natural helical branch only.

7. Absolute Pointwise Reconstruction

There is a second formulation that should not be confused with the incremental model. It uses an absolute branch counter $M(k)$ rather than an incremental counter $m(n)$. Define

$$\gamma_{\mathrm{phase,arg}}(k) = \frac{1}{\alpha}\left(\omega + \mathrm{Arg}_{\mathrm{lift}}\left(\omega + i\frac{B}{k}\right)\right). \tag{7.1}$$

In computation this is evaluated with

$$\gamma_{\mathrm{phase,arg}}(k) = \frac{\omega + \mathrm{atan2}(B/k, \omega)}{\alpha}, \tag{7.2}$$

together with a pointwise integer branch choice. The absolute reconstruction is

$$\gamma_{\mathrm{pred,arg}}(k) = \gamma_{\mathrm{phase,arg}}(k) + \frac{2\pi}{\alpha} M(k). \tag{7.3}$$

For the reported diagnostic reconstruction, the absolute counter was chosen as

$$M(k) = \mathrm{round}\left(\frac{\gamma_{\mathrm{real}}(k) - \gamma_{\mathrm{phase,arg}}(k)}{2\pi/\alpha} \right). \tag{7.4}$$

Warning 7.1. The symbols $m(n)$ and $M(k)$ denote different objects. The former is an incremental branch count for a gap. The latter is an absolute accumulated branch count at a point. Mixing them leads to an incorrect formulation.

8. The Logarithmic Term Generated by the Helix

This section records a heuristic point that is important for interpreting the fit. The logarithmic term is not inserted through the Riemann–von Mangoldt formula. It is generated by the radial derivative of the helical ansatz itself.

Heuristic 8.1. Consider the centered helix

$$Z(k) = H(k) - C = Ak^B \exp(i(\omega k + \theta_0)). \tag{8.1}$$

Its derivative is

$$Z'(k) = Ak^B \exp(i(\omega k + \theta_0)) \left(\frac{B}{k} + i\omega \right). \tag{8.2}$$

Thus the tangent direction contains the argument of

$$\frac{B}{k} + i\omega, \tag{8.3}$$

or, equivalently after a fixed rotation convention, the argument of

$$\omega + i\frac{B}{k}. \tag{8.4}$$

For large k, on a fixed lifted branch,

$$\mathrm{Arg}_{\mathrm{lift}} \left(\omega + i\frac{B}{k} \right) = \phi_\infty + \frac{B}{\omega k} + O\left(\frac{1}{k^2} \right), \tag{8.5}$$

where ϕ_∞ is the limiting lifted argument. Substitution into (3.4) gives an expansion of the form

$$\Delta_h(k) = a + \frac{b}{k} + O\left(\frac{1}{k^2} \right), \tag{8.6}$$

with constants a and b depending on α, B, ω, and the chosen lifted branch.

Summing (8.6) gives

$$\sum_{k=1}^{N} \Delta_h(k) = aN + b \sum_{k=1}^{N} \frac{1}{k} + O(1) \tag{8.7}$$

$$= aN + b \log N + O(1). \tag{8.8}$$

The logarithm in (8.8) is therefore a consequence of the relative radial derivative

$$\frac{(k^B)'}{k^B} = \frac{B}{k}. \tag{8.9}$$

Remark 8.2. Equation (8.8) produces a logarithm in the index N, not directly a logarithm in the height T. The Riemann–von Mangoldt counting formula has a logarithmic dependence on the height variable T; see the standard references [2, 3]. The bridge between these two logarithms is the asymptotic relation between index and height:

$$N(T) \sim \frac{T}{2\pi} \log T. \tag{8.10}$$

Consequently,

$$\log N = \log T + \log\log T - \log(2\pi) + o(1). \tag{8.11}$$

Thus the helical model naturally generates a $\log N$ correction; after passing from index to height, this contains a $\log T$ component. This is a compatibility statement, not a derivation of the Riemann–von Mangoldt formula from the fit.

8.1. **Changing from index to height.** The helical curve in (2.1) is a curve in the index variable k:

$$H_k = C + Ak^B \exp(i(\omega k + \theta_0)). \tag{8.12}$$

The zero-counting function, however, is naturally expressed in a height variable T. The correct comparison is therefore not between ωk and T, but between k and $N(T)$.

If the index is written as a function of height, the exact change of variable is formally

$$H(T) = C + AN(T)^B \exp(i(\omega N(T) + \theta_0)). \tag{8.13}$$

For a smooth Riemann–von Mangoldt type approximation, replace $N(T)$ by its main smooth term

$$F(T) = \frac{T}{2\pi} \log\left(\frac{T}{2\pi e}\right) + \frac{7}{8}. \tag{8.14}$$

This gives the height-parametrized helical thread

$$H_{\mathrm{RvM}}(T) = C + AF(T)^B \exp(i(\omega F(T) + \theta_0)). \tag{8.15}$$

In this variable the phase is not linear in T:

$$\Theta(T) = \omega F(T) + \theta_0. \tag{8.16}$$

Its derivative is

$$\Theta'(T) = \omega F'(T) = \frac{\omega}{2\pi} \log\left(\frac{T}{2\pi}\right). \tag{8.17}$$

Thus the angular velocity in height grows like $\log T$.

On the other hand, the local spacing between consecutive zeros is heuristically the reciprocal of the density:

$$\gamma_{k+1} - \gamma_k \approx \frac{1}{F'(\gamma_k)} = \frac{2\pi}{\log(\gamma_k/2\pi)}. \tag{8.18}$$

Therefore the angular increment per zero is

$$\Theta(\gamma_{k+1}) - \Theta(\gamma_k) \approx \Theta'(\gamma_k)(\gamma_{k+1} - \gamma_k) \tag{8.19}$$

$$\approx \frac{\omega}{2\pi} \log\left(\frac{\gamma_k}{2\pi}\right) \cdot \frac{2\pi}{\log(\gamma_k/2\pi)} \tag{8.20}$$

$$= \omega. \tag{8.21}$$

This is the key compatibility:

$$\text{linear phase in } k \quad \Longleftrightarrow \quad \text{phase } \omega N(T) \text{ in height } T. \tag{8.22}$$

The derivative of the height-parametrized thread is

$$H'_{\mathrm{RvM}}(T) = \big(H_{\mathrm{RvM}}(T) - C\big)\left(\frac{BF'(T)}{F(T)} + i\omega F'(T)\right). \tag{8.23}$$

Since

$$F'(T) = \frac{1}{2\pi}\log\left(\frac{T}{2\pi}\right), \tag{8.24}$$

this becomes

$$H'_{\mathrm{RvM}}(T) = \big(H_{\mathrm{RvM}}(T) - C\big)\frac{1}{2\pi}\log\left(\frac{T}{2\pi}\right)\left(\frac{B}{F(T)} + i\omega\right). \tag{8.25}$$

Consequently,

$$\big|H'_{\mathrm{RvM}}(T)\big| = AF(T)^B\frac{1}{2\pi}\log\left(\frac{T}{2\pi}\right)\sqrt{\left(\frac{B}{F(T)}\right)^2 + \omega^2}. \tag{8.26}$$

If $\omega \neq 0$, then as $T \to \infty$,

$$\big|H'_{\mathrm{RvM}}(T)\big| \sim \frac{A|\omega|}{2\pi}F(T)^B\log\left(\frac{T}{2\pi}\right). \tag{8.27}$$

Remark 8.3. The error in comparing the helical thread directly with a height variable is to compare ωk with T. The helix is linear in the index variable: ωk. In height it becomes $\omega N(T)$, and, using the smooth counting approximation, its phase is approximately

$$\Theta(T) \sim \frac{\omega T}{2\pi}\log\left(\frac{T}{2\pi e}\right). \tag{8.28}$$

Thus a static helical thread in k becomes a Riemann–von Mangoldt shaped thread in the height variable T.

8.2. **The height equation with branches.** The smooth height-parametrized curve (8.15) is useful for describing the continuous thread, but it is not sufficient for reconstructing individual zero ordinates. To recover absolute ordinates one must keep the integer branch data. A branch-aware height equation has the form

$$\alpha T = \omega F(T) + \theta_0 + \mathrm{Arg}_{\mathrm{lift}}\left(\omega + i\frac{B}{F(T)}\right) + 2\pi M(T), \tag{8.29}$$

where $M(T)$ is an integer branch counter. Equivalently, for the k-th zero one may write

$$\alpha\gamma_k = \omega k + \theta_0 + \mathrm{Arg}_{\mathrm{lift}}\left(\omega + i\frac{B}{k}\right) + 2\pi M_k. \tag{8.30}$$

The formula without $M(T)$ or M_k only describes a phase modulo 2π; it does not determine the absolute height.

For the present value $\alpha = 3.43397596$, the provoked branch excesses are encoded by a degree-16 Newton-binomial model modulo

$$p = 1{,}000{,}003. \tag{8.31}$$

For $r = 1, \ldots, 17$,

$$j(r) \equiv \sum_{q=0}^{16} A_q \binom{r-1}{q} \pmod{p}, \tag{8.32}$$

$$\eta(r) \equiv \sum_{q=0}^{16} B_q \binom{r-1}{q} \pmod{p}. \tag{8.33}$$

Here $j(r)$ is the gap where the provoked jump occurs and $\eta(r) = m(j(r)) - 1$ is the excess over the natural branch continuation. The coefficients are shown in Table 2.

TABLE 2. The 17 modular coefficients for the provoked branch events at $\alpha = 3.43397596$.

q	A_q	B_q
0	1	3
1	1	1000001
2	0	3
3	1	999998
4	1000000	8
5	7	999991
6	999990	17
7	20	999980
8	999978	30
9	20	999965
10	15	47
11	999881	999946
12	370	68
13	999164	999923
14	1577	93
15	997400	999896
16	4311	122

The verification is exact:

$$\max_{1\le r\le 17} |j_{\mathrm{model}}(r) - j_{\mathrm{data}}(r)| = 0, \tag{8.34}$$

$$\max_{1\le r\le 17} |\eta_{\mathrm{model}}(r) - \eta_{\mathrm{data}}(r)| = 0. \tag{8.35}$$

The corresponding total jump multiplier is

$$m(j(r)) = 1 + \eta(r). \tag{8.36}$$

9. Numerical Error Comparisons

The original helical reconstruction using the scalar arctangent produced, at the last zero,

$$\gamma_{\mathrm{real}}(N) = 1132490.658714411081746, \tag{9.1}$$

$$\gamma_{\mathrm{pred}}(N) = 1132490.784989385900000, \tag{9.2}$$

$$\gamma_{\mathrm{pred}}(N) - \gamma_{\mathrm{real}}(N) = 0.126274974783883. \tag{9.3}$$

Since

$$\frac{\pi}{\alpha} = 0.9148557503558624, \tag{9.4}$$

the last-point error is not explained by adding a half-turn branch correction:

$$0.126274974783883 + \frac{\pi}{\alpha} = 1.041130725139745. \tag{9.5}$$

The pointwise $\mathrm{Arg}_{\mathrm{lift}}$ reconstruction (7.3)–(7.4) gives

$$\gamma_{\mathrm{pred,arg}}(N) = 1132489.870133635600000, \tag{9.6}$$

$$\gamma_{\mathrm{pred,arg}}(N) - \gamma_{\mathrm{real}}(N) = -0.788580775493756. \tag{9.7}$$

The global error statistics for the old scalar-arctangent reconstruction were

$$\mathrm{MAE}_{\mathrm{old}} = 0.457638591251554, \tag{9.8}$$

$$\mathrm{RMSE}_{\mathrm{old}} = 0.528369468968370, \tag{9.9}$$

$$\max|\mathrm{error}_{\mathrm{old}}| = 0.914855492534116. \tag{9.10}$$

For the pointwise $\mathrm{Arg}_{\mathrm{lift}}$ reconstruction, the statistics were

$$\mathrm{MAE}_{\mathrm{arg}} = 0.457217159104298, \tag{9.11}$$

$$\mathrm{RMSE}_{\mathrm{arg}} = 0.528004494407613, \tag{9.12}$$

$$\max|\mathrm{error}_{\mathrm{arg}}| = 0.914855594746769. \tag{9.13}$$

The height-variable laboratory described in (8.29) was then run with the full-set helical fit. It compared three reconstructions: the smooth inverse of $F(T) = k$, the branch-aware height helix with branches read diagnostically from the real zeros, and the branch-aware height helix with branches seeded by the smooth Riemann–von Mangoldt inverse. At the last zero the errors were

$$\mathrm{error}_{F^{-1}}(N) = 0.323464922374114, \tag{9.14}$$

$$\mathrm{error}_{\mathrm{helix,real\ branch}}(N) = 0.485718277748674, \tag{9.15}$$

$$\mathrm{error}_{\mathrm{helix,RvM\ branch}}(N) = 0.485718277748674. \tag{9.16}$$

The corresponding global statistics were

$$\mathrm{MAE}_{F^{-1}} = 0.293088963274172, \tag{9.17}$$

$$\mathrm{RMSE}_{F^{-1}} = 0.341034082010784, \tag{9.18}$$

$$\max|\mathrm{error}_{F^{-1}}| = 3.713111370849617, \tag{9.19}$$

for the smooth inverse, while the branch-aware height helix with diagnostic real branches gave

$$\mathrm{MAE}_{\mathrm{helix,real\ branch}} = 0.457389526843916, \tag{9.20}$$

$$\mathrm{RMSE}_{\mathrm{helix,real\ branch}} = 0.528946892389236, \tag{9.21}$$

$$\max|\mathrm{error}_{\mathrm{helix,real\ branch}}| = 0.914852803456597. \tag{9.22}$$

With branches seeded by the smooth inverse, the height helix gave

$$\mathrm{MAE}_{\mathrm{helix,RvM\ branch}} = 0.527712910326030, \tag{9.23}$$

$$\mathrm{RMSE}_{\mathrm{helix,RvM\ branch}} = 0.639124598835634, \tag{9.24}$$

$$\max|\mathrm{error}_{\mathrm{helix,RvM\ branch}}| = 3.231018912401119. \tag{9.25}$$

Thus, in this height-variable laboratory, the smooth inverse has the smaller last-zero error and the smaller mean errors, whereas the branch-aware helical model with diagnostic branches has the smaller maximum error over the tested range.

10. Conclusion

The main conceptual correction is the replacement of a monovalued arctangent by a lifted complex argument. The helical phase should be written as

$$\mathrm{Arg}_{\mathrm{lift}}\left(\omega + i\frac{B}{n}\right) \tag{10.1}$$

or, in the absolute formulation, as

$$\mathrm{Arg}_{\mathrm{lift}}\left(\omega + i\frac{B}{k}\right). \tag{10.2}$$

The incremental formulation uses $m(n)$, while the pointwise formulation uses $M(k)$. These counters encode different branch data and should be kept separate.

Remark 10.1. For $\alpha = 3.43397596$, the modular table contains the 17 useful provoked branch events satisfying $m(n) > 1$. The last occurs at $n = 126$. After that index, the model has no provoked jump term and proceeds with the natural helical branch.

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